CFD-HW6

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Outline

In this assignment, heat and pressure dynamics will be modeled during spring break across a heat exchanger with periodic geometry. To be specific, to cases of the geometry will be analyzed as prescribed.

Preliminary Calculations

The (effective) inlet fluid velocity per periodic module is give by:

$$V = \frac{\dot{m}}{\rho H L} = \frac{0.05}{998.2 \cdot (0.01) \cdot (1)} \mathrm{m/s} \approx 0.005\,\mathrm{m/s}$$

The Hydraulic diameter can be computed as such

$$D_h = \frac{2HL}{H+L} = \frac{2(0.01)(1)}{0.01+1} \mathrm{m} \approx 0.02\,\mathrm{m}$$

The Reynolds number can be approximated as such:

$$\mathrm{Re} \approx \frac{\rho V D_h}{\mu} \approx \frac{(998.2)(0.005)(0.02)}{0.001} \approx 100 \implies \text{Laminar Flow}$$

The pressure drop in general could thus be approximated as:

$$\Delta P \approx \frac{64}{\mathrm{Re}} \frac{L_{flow}}{D_h} \frac{\rho V^2}{2}$$

For case 1:

$$L_{flow} = 2(20\,\mathrm{mm}) = 0.04\,\mathrm{m} \implies \Delta P_1 \approx \frac{64}{100} \frac{0.04}{0.02} \frac{(998.2)(0.005)^2}{2} \approx 0.016\,\mathrm{Pa}$$

For case 2:

$$L_{flow} = 2(30\,\mathrm{mm}) = 0.06\,\mathrm{m} \implies \Delta P_2 \approx \frac{64}{100} \frac{0.06}{0.02} \frac{(998.2)(0.005)^2}{2} \approx 0.025\,\mathrm{Pa}$$

For convenience, the primary mode of heat transfer will be assumed to be conduction. The heat transfer rate in general is given by:

$$\dot{Q} = kA\frac{T_{wall} - T_{bulk}}{\delta x} \quad \text{where} \quad A = 2(\pi D L)$$

For case 1:

$$D = 0.01\,\mathrm{m},\ \delta x \approx H/2 = 0.005\,\mathrm{m} \implies \dot{Q}_1 \approx (0.6)\,(2(\pi(0.01)))\,\frac{400-300}{0.005} \approx 754\,\mathrm{W}$$

For case 2:

$$D = 0.0075\,\mathrm{m},\ \delta x \approx H/2 = 0.005\,\mathrm{m} \implies \dot{Q}_2 \approx (0.6)\,(2(\pi(0.0075)))\,\frac{400-300}{0.005} \approx 565\,\mathrm{W}$$

Upon noting that in general one has $\Delta T = \dot{Q}/(\dot{m}c_p)$:

$$\text{For case 1:} \quad \Delta T_1 \approx \frac{754}{(0.05)(4182)}\mathrm{K} \approx 3.6\,\mathrm{K} \quad \text{and} \quad \text{for case 2:} \quad \Delta T_2 \approx \frac{565}{(0.05)(4182)}\mathrm{K} \approx 2.7\,\mathrm{K}$$

Geometry

Because we were so graciously not given the geometry files, I put them together in Ansys Discovery as follows with only a minor amount of pain:

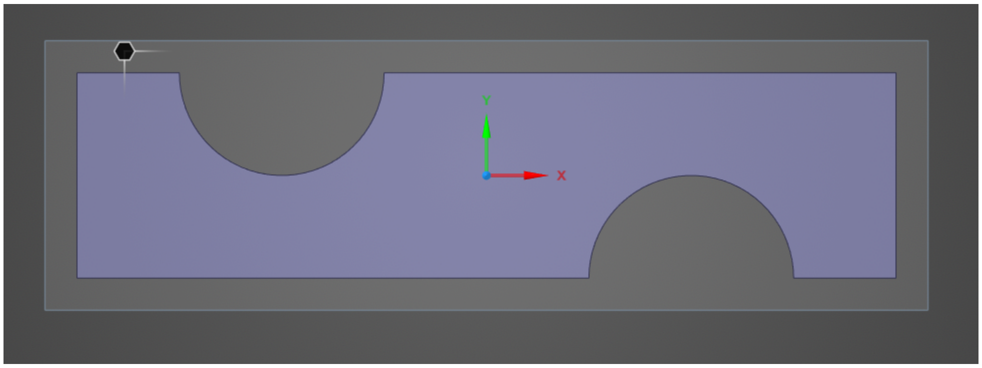

Figure 1.1: Preliminary Geometry Work (Case 1)

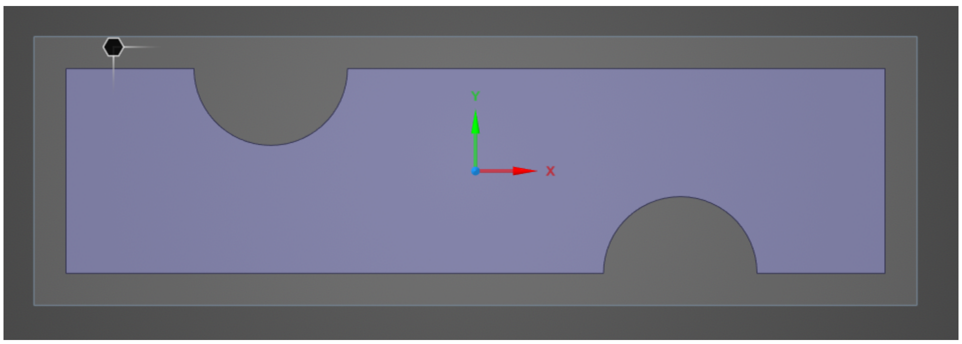

Figure 1.2: Preliminary Geometry Work (Case 2)

Meshing

Using the inflation tool on the circular arc boundaries and a generic cell size of 0.0001 m (quad), I was able to achieve the following meshes:

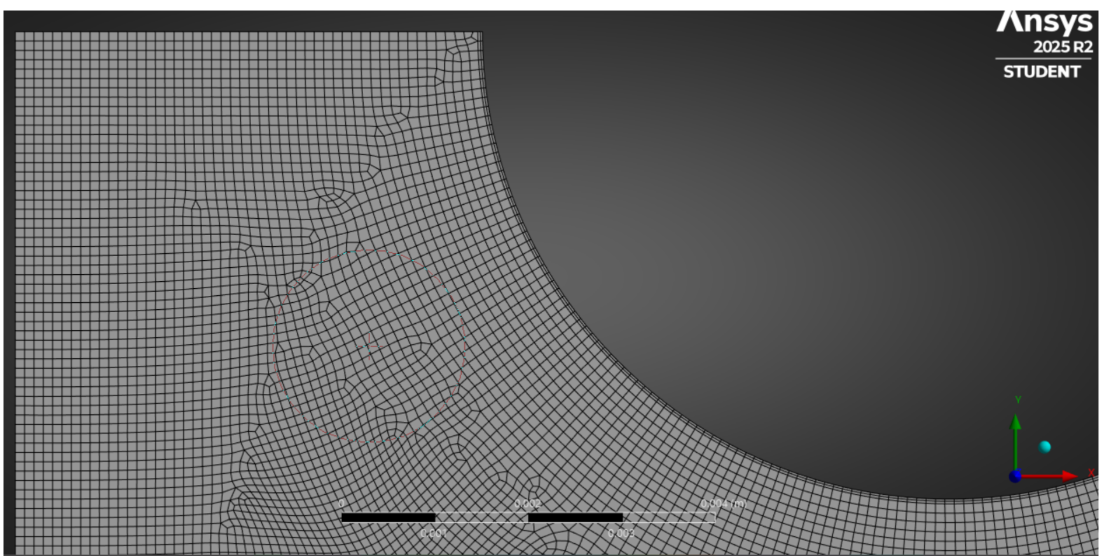

Figure 2.1: Partial Mesh (Case 1)

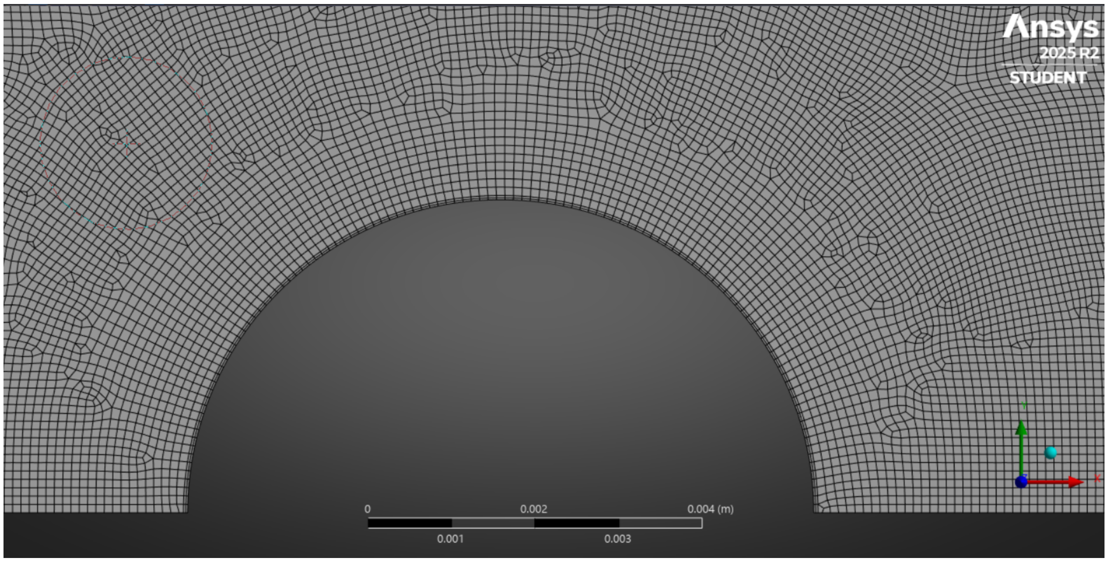

Figure 2.2: Partial Mesh (Case 2)

Solution - Case 1

By following the outlined procedure, the periodic boundary conditions were applied accordingly and the following residuals were achieved after running the simulation:

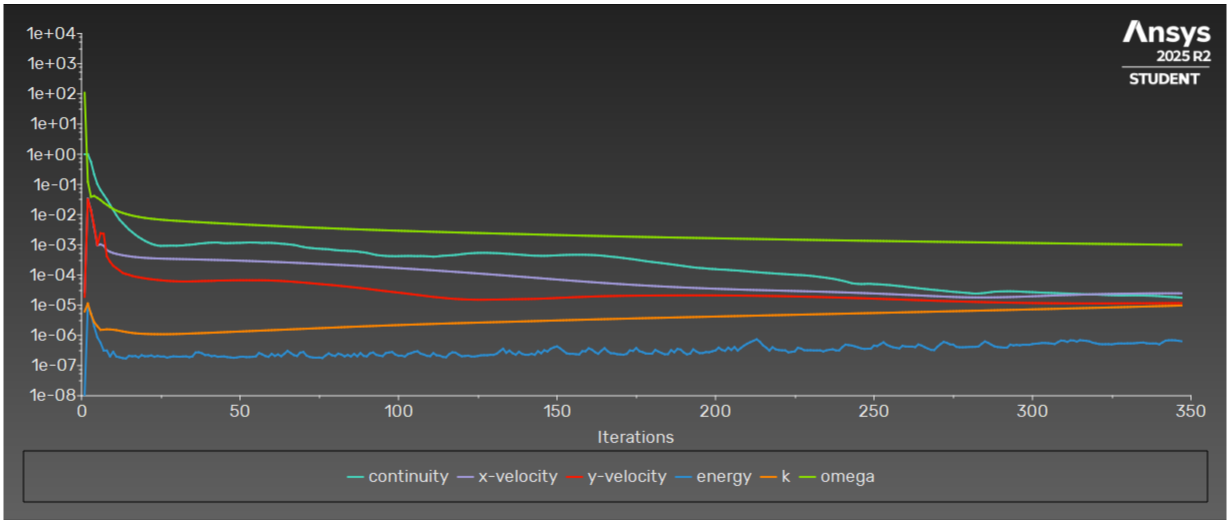

Figure 3.1: Residuals (Case 1)

The energy residual is just under 10^{-6}, so we're swimming in some okay water. Not great, but not horrific.

Regarding the results, the average pressure at the inlet was determined to be 4.12 Pa, and the average pressure at the outlet was determined to be 4.12 Pa strangely, implying a pressure drop of 0. This doesn't exactly line up with the predicted pressure drop, but it is on the same order of magnitude. Here is the pressure contour for reference:

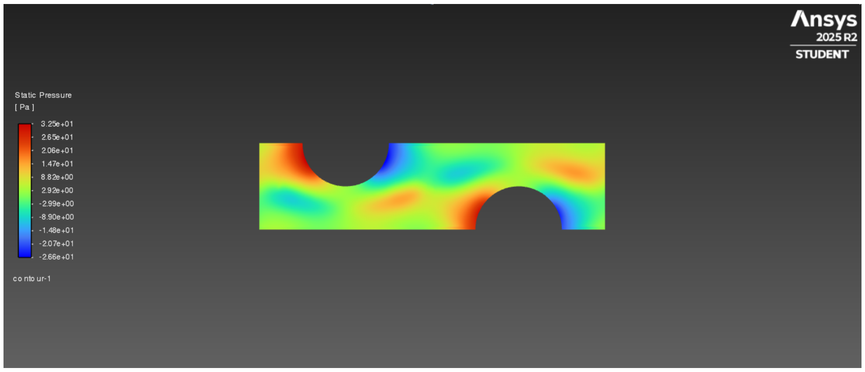

Figure 3.2: Pressure (Case 1)

The speed field is presented here as well:

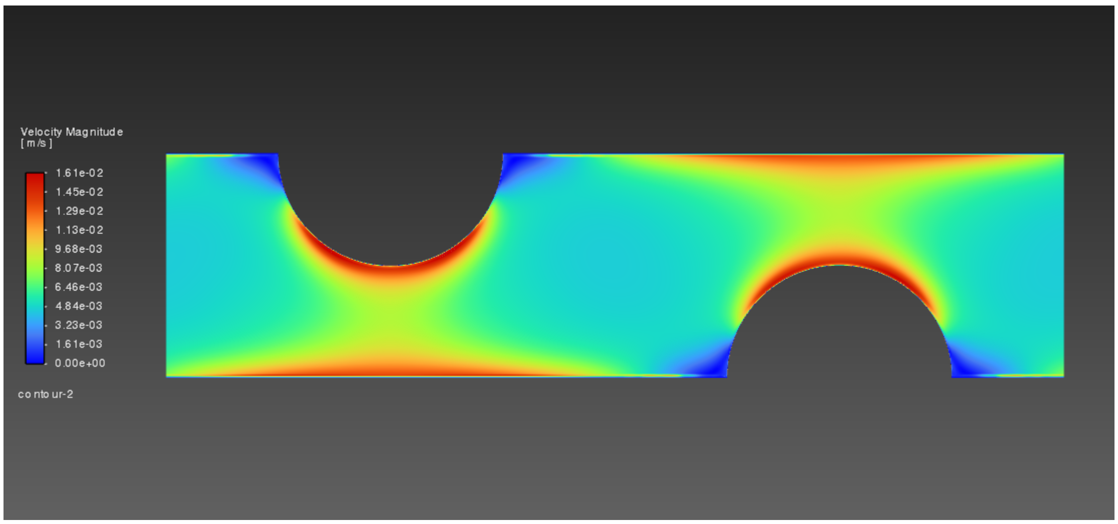

Figure 3.3: Speed (Case 1)

Next up is temperature:

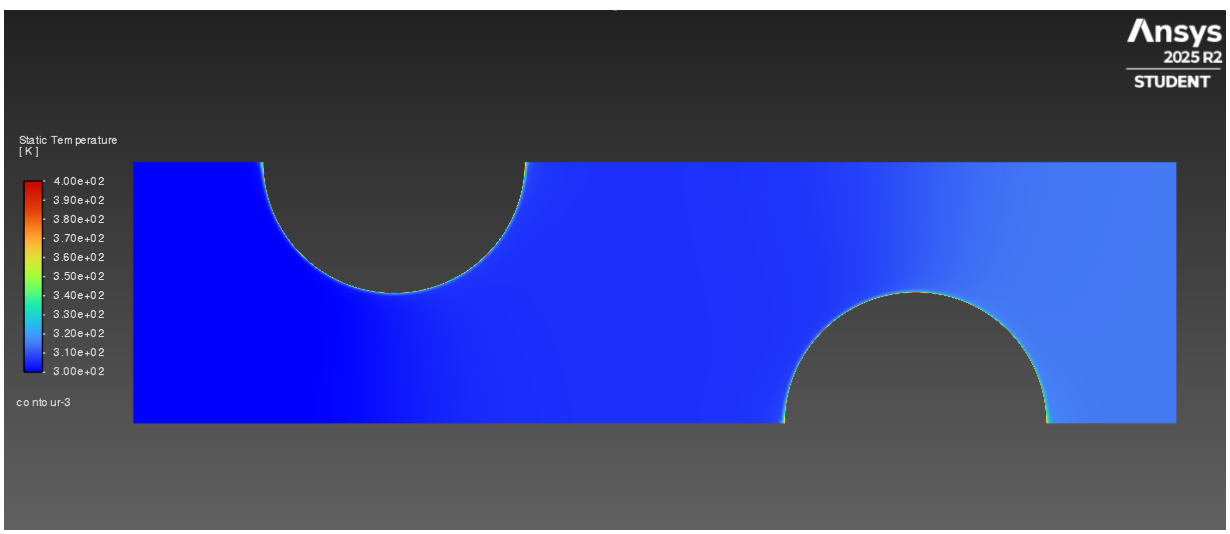

Figure 3.4: Temperature (Case 1)

Something is pretty off/boring here. But the temperature drop across the section does appear to be on the order of 1-10 K, which is what was predicted, which is strange.

Solution Case 2

Once again, the periodic BCs were implemented and the residuals came out to this:

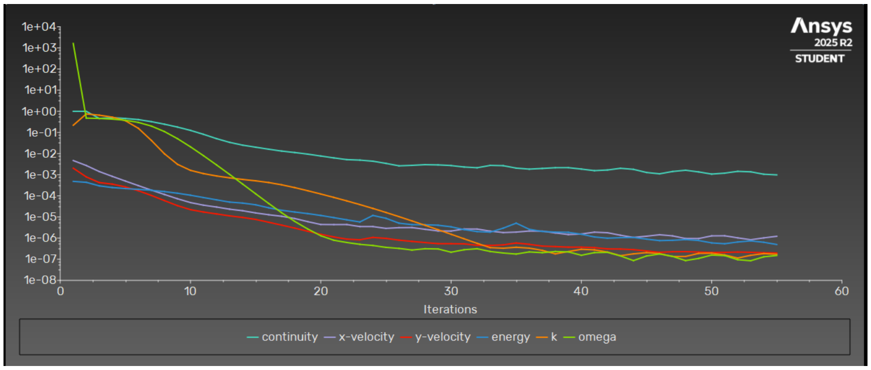

Figure 4.1: Residuals (Case 2)

And much to my surprise, the energy residual behaved very similarly to before, but it converged faster for some reason.

Regarding pressure, the pressure distribution was determined to be as such:

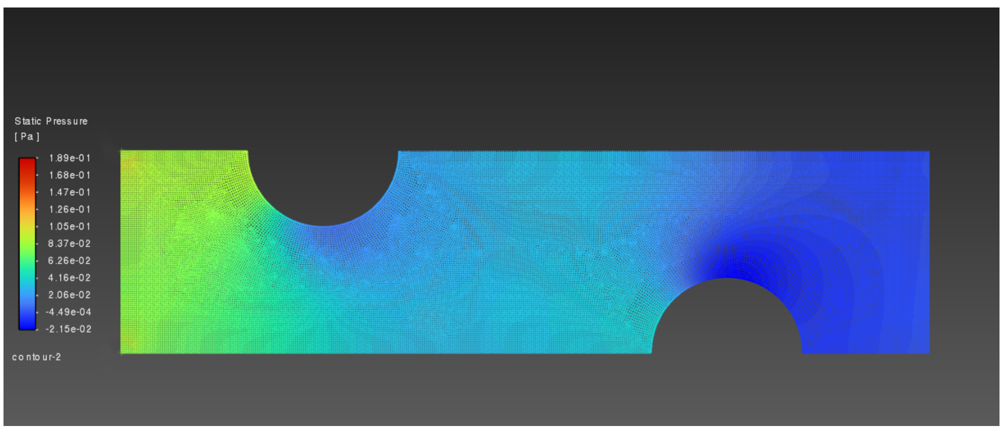

Figure 4.2: Pressure (Case 2)

I'm definitely doing something wrong... I think the tutorial is sabotaging me. This pressure gradient is not only off from expectations, it's also off from the previous result. Anyway, let's visit velocity!

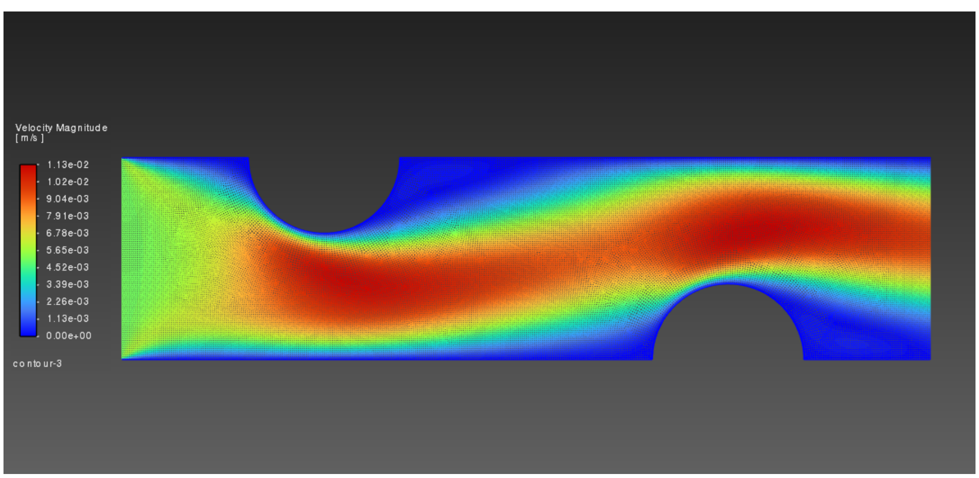

Figure 4.3: Speed (Case 2)

I don't know what's going on, here. It's very possible I somehow screwed up the periodic boundary conditions. It almost looks as though the periodic wall boundaries are standard walls, even though everything was (apparently) set up correctly. Really strange. Moving on to temperature.

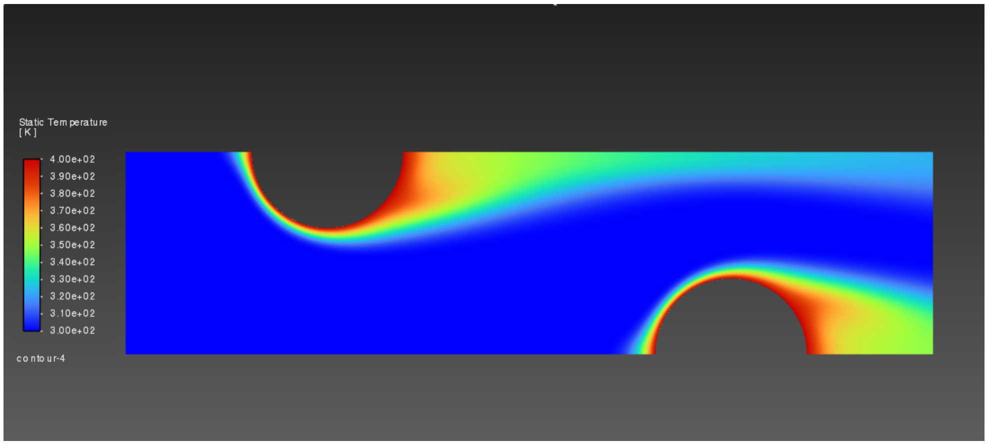

Figure 4.4: Temperature (Case 2)

Yeah... I'm pretty confident the periodic boundary conditions aren't working at all. That's quite unfortunate. The temperature change does not line up, and I expected that to occur because I neglect convection. Otherwise, it seems reasonable.

Discussion

Overall, the periodic boundary conditions were not adequately implemented potentially because I don't know how to follow a tutorial, and I'm trying to get this done in under an hour and a half, or because something is terribly wrong with my Fluent settings. Not sure to be honest.

More quantitatively, the velocity predictions were more or less correct, the pressure predictions were all over the place, and let's not talk about temperature... though that's because convection is likely dominant here, but I modeled this as conduction dominant likely because I saw k, and jumped the gun.

In conclusion, this was a disaster, but there's only so much time in the world. I'll likely revisit this assignment on my own time to make sure my screw ups are conceptual, but this is what I can manage to crank out in a time crunch.